FilaSeg: Geometry-based instance reconstruction of carbon nanotube bundles from degraded centreline masks

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Abstract

Disentangling dense filamentary networks into individual instances is challenging because gaps interrupt true trajectories, while crossings create several locally plausible continuations. We present FilaSeg, a non-learning, geometry-based method that takes a binary filament-axis mask and separates it into individual filament instances through residual vectorization, fragment cleaning, and staged, topology-checked geometric reconnection. Its instance grouping performance is evaluated on procedurally generated filament networks with exact ground truth and compared with a minimum-turn skeleton tracer. Methods are assessed using a common, method-independent partition of the input skeleton into fragments, with pairwise grouping F_1 calculated on equivalent per-instance centreline representations before width rendering.

After the method settings and evaluation protocol were fixed, FilaSeg and the minimum-turn tracer were evaluated on 125 independently seeded procedural scenes. FilaSeg achieved a mean common-fragment F_1=0.734 (density-stratified 95% bootstrap CI [0.722, 0.745]), with a paired FilaSeg-minus-minimum-turn difference of 0.071 (95% CI [0.059, 0.082]). The advantage was not uniform across density: it was resolved at four of the five tested densities, while the interval crossed zero at the fifth. Re-scoring the same scenes on undegraded masks narrowed the paired difference to 0.042 [0.031, 0.053], indicating that much of the aggregate advantage reflects insensitivity to mask degradation rather than crossing disambiguation alone.

FilaSeg was subsequently applied to scanning electron microscopy (SEM) images of carbon-nanotube (CNT) sheets. For this application, a separate upstream CycleGAN/U-Net workflow converts the SEM data into binary masks representing the axes of the CNT bundles. These masks are supplied to FilaSeg, while the paired SEM image is used again in Stage 4 to estimate apparent bundle widths, handle overlapping rendered instances, and produce the final output. Evaluating the rendered Stage 4 output instead of the Stage 3 centreline representation produced a lower grouping score, indicating that centreline grouping should be assessed separately from width rendering and final-output postprocessing.

A single manually annotated SEM crop was used during development and is not a held-out test. The predefined synthetic evaluation therefore supports a conditional advantage over the fixed minimum-turn comparator for the tested generator.

Note: A broader upstream pipeline also exists around the method described in this manuscript: real SEM images and procedurally generated masks → CycleGAN image translation → U-Net semantic segmentation → binary CNT bundle-axis mask → FilaSeg instance extraction → SEM-assisted Stage-4 width estimation, overlap handling, and rendering. This manuscript currently treats the binary axis mask as the formal input and evaluates only the instance-extraction stages. To discuss:

- Should the complete upstream CycleGAN/U-Net pipeline be integrated into this paper?

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- Should it instead remain a separate paper or companion workflow?
- Should the current manuscript continue to treat the binary axis mask as the formal input?
- Give Stage 4 (SEM-assisted measurement/postprocessing) its own discussion.

Keywords: carbon nanotubes; scanning electron microscopy; instance segmentation; curvilinear structure extraction; Hough transform; filament tracing; image-based microstructural characterization.

Review note: A central framing question for this manuscript is how the Introduction should be organized. Should the paper begin with the general problem of instance segmentation and disentanglement in dense filamentary-network images, with CNT bundle sheets presented as the main application and validation case? Alternatively, should the Introduction begin with the need to characterize CNT sheets from SEM images, with FilaSeg presented primarily as a CNT-specific analysis method?

A possible general-to-specific structure is:

1. Introduce the general difficulty of separating individual filaments in dense, crossing, and fragmented networks.
2. Explain why conventional connected-component methods and compact-object instance-segmentation approaches are poorly suited to long, thin, intersecting structures.
3. Introduce CNT bundle sheets as an important application for which individual bundle geometry must be recovered from SEM images.
4. Describe how the upstream CycleGAN/U-Net workflow produces the binary CNT bundle-axis mask supplied to FilaSeg.
5. Review relevant filament-tracing, learned instance-segmentation, and CNT-image-analysis methods.
6. Identify the methodological gap addressed by FilaSeg.
7. State the contribution, evaluation strategy, and principal limitations.

The alternative CNT-first structure would begin with the scientific need to measure individual bundles in CNT sheets and would then introduce dense filament disentanglement as the image-analysis problem preventing this.

Which framing is more appropriate: a general filament-instance-segmentation paper demonstrated on CNT SEM images, or a CNT-sheet characterization paper centred on the complete SEM-to-instance workflow?

1 Introduction

The macroscopic properties of carbon nanotube (CNT) films, yarns and composites are governed by the geometry of the bundle network rather than by the properties of individual nanotubes alone: the length distribution of the bundles, their width, their local orientation and alignment, their curvature, and the density and topology of the crossings between them all enter the mechanical and electrical response [1, 2]. Mesoscopic and network models of CNT assemblies take exactly this description, slender mutually contacting elements with measurable geometry, as their input [3–5]. Scanning electron microscopy (SEM) is the routine imaging modality for such materials, yet the quantitative analysis that follows an acquisition usually stops at bulk descriptors: an areal coverage fraction, a mean width, or a global orientation-order parameter.

These summarize the network but discard the object that quantitative characterization actually needs, the individual bundle as an addressable instance with its own centreline, width, endpoints and crossings. Image analysis of CNT micrographs has followed the same emphasis, quantifying nematic order or the orientation distribution of CNT forests and films [6, 7]. Recent work has also applied neural networks to identify CNT foreground in microscopy images [8, 9], while broader reviews summarize deep-learning segmentation across nanomaterial imaging [10].

The obstacle is not semantic segmentation. Convolutional segmentation networks such as U-Net [11] and its self-configuring successors [12] produce good foreground masks of CNT structures. Instance-level extraction, partitioning that foreground into the correct set of individual bundles, remains challenging in dense, overlapping filament networks, for four coupled reasons. First, bundles are effectively one-dimensional objects embedded in a two-dimensional image, so a single faint patch a few pixels long splits one bundle into two mask fragments. Second, bundles cross and touch, so foreground connectivity does not correspond to bundle identity: two crossing bundles form one connected component that must be split, while one faded bundle forms two components that must be joined. Third, the bundles are densely packed, so at a crossing a locally correct continuation and a locally plausible wrong continuation are often geometrically indistinguishable. Fourth, the objects are only a few pixels wide, so pixel-overlap instance metrics and one-pixel skeleton artefacts dominate naive approaches. Standard instance-segmentation priors assume a compact object with a clear interior and boundary: a bounded blob, a flow field pointing toward a centre, or a star-convex shape [13–15]. CNT bundles do not fit that assumption. They are long, thin, crossing curves with no meaningful centre, and they may overlap.

Four classes of existing method address parts of this problem. *Ridge and tubular-structure detectors* [16–18] recover sub-pixel line positions and enhance line-like structure from grayscale intensity, but they do not assign pixels to individual objects across a junction. *Fibre-network tracing tools* developed for collagen and biopolymer networks, including FIRE and CT-FIRE [19, 20], SOAX [21] and the FiNTA tracing algorithm [22], return per-fibre length, width and orientation, but they target sparser, often micron-scale fibres and yield a fibre graph rather than a set of separately rendered, individually addressable instances. *Aggregate measurement tools* such as DiameterJ [23] deliver well-validated width and pore-size distributions over a whole micrograph, but as image-level distributions rather than per-instance geometry, and they do not resolve crossing ownership; the individual-fibre separation approach of Depriester et al. [24] addresses that separation step directly for fibrous materials. *Learned instance methods* for filaments are emerging: FibeR-CNN [25] extends Mask R-CNN to image-based fibre analysis, including CNT-like fibres, as an independent raw-image detector rather than a centreline-mask grouping method, and Pick and Trace [26] segments filamentous instances with a recurrent network. Methodologically closest to the present reconnection stage is the terminus-pairing approach of Liu et al. [27], which separates a raw microscopy image into orientation-specific branches with a learned network and then pairs terminus points across branches to regroup fragments into individual filaments; it was validated on microtubule micrographs rather than on a precomputed centreline mask. That two-step strategy of first separating the network by local orientation and then reconnecting fragments by pairing their endpoints is the same decomposition FilaSeg applies, although the two methods realise it differently: Liu et al. [27] obtain the orientation separation from a trained network operating on the grayscale image, whereas the orientation binning and endpoint pairing used here are explicit geometric criteria applied to a binary axis mask, with no learned component. We therefore regard that work as the closest published methodological relative of the reconnection stage rather than as a component this method builds on. Instance tracing of crossing curvilinear objects is also mature in other domains, in neuron reconstruction [28–30], plant-root architecture [31] and crack segmentation [32], but each embeds a domain-specific prior that dense CNT bundle films or networks (referred to here as *CNT mats*) do not satisfy: bundles are open curves that cross freely and carry neither a root nor a hierarchy.

The closest published work is that of Peters et al. [33], who combine U-Net-like semantic segmentation with algorithmic fibre tracing, crossing separation and width refinement from transverse grayscale profiles, and validate on SEM images of CNT materials for aerosol characterization and exposure assessment. Two-stage semantic-segmentation-plus-tracing designs therefore already exist for CNT SEM, and the contribution claimed here is not the two-stage idea itself. What distinguishes the present work is the operating regime and the representation. Peters et al. [33] target low-occupancy aerosol-filter images, in which fibres are comparatively isolated; the target here is a dense CNT mat in which the crossing density, not the detectability of a single fibre, sets the difficulty. The input representation differs accordingly: FilaSeg consumes a thin-axis centreline mask rather than a full bundle silhouette, and emits a multi-label representation in which overlapping instances retain their shared pixels at a crossing instead of one instance claiming them. The method differs in using tiled Hough-based residual vectorization followed by staged, topology-checked reconnection, and the evaluation differs in reporting a pair-weighted grouping diagnostic that separates over-merging from under-merging, in stress-testing across a controlled range of crossing density, and in adapting every pixel-scale parameter to the physical pixel size. These are differences in design, not a head-to-head performance comparison with Peters et al. [33] or the other related methods above; the present evidence does not establish quantitative superiority over them (Section 4.2).

This paper presents FilaSeg, a geometry-based postprocessing method that converts a binary filament-centreline mask and its paired SEM image into separated filament instances. The input is a binary centreline mask together with the grayscale SEM image from which it was derived; the output is a set of possibly overlapping instances, each carrying a pixel support, an ordered centreline, a width, a length and endpoints, stored as a multi-label stack so that crossings are preserved. These geometric fields are exported for downstream use but are not claimed as validated measurements in the present study. The method proceeds in four stages: tiled Hough-based residual vectorization, orientation-aware fragment cleaning, geometry-based instance reconnection, and SEM-gradient-based width estimation with final rendering. It is evaluated on one manually annotated real SEM crop and on a procedural synthetic benchmark that renders thick bundles for the SEM channel while feeding the method the thin axis that an upstream segmenter actually reconstructs. Three questions are addressed: how grouping accuracy depends on the generator's target areal density; whether the conclusion changes when methods are scored on equivalent centreline representations; and which scenes favour FilaSeg or a minimum-turn comparator. The semantic segmentation that produces the input mask is treated throughout as an external upstream process and is not evaluated here. The real-data evidence rests on a single annotated development crop, so it is presented qualitatively rather than as a population estimate.

2 Materials and methods

This section describes the two connected image-analysis pipelines used in this work, the synthetic and real input data, the four stages of FilaSeg, and the procedure used to evaluate its instance-segmentation performance.

The upstream binary-segmentation pipeline converts an SEM image into a binary mask representing the axes of the CNT bundles. FilaSeg then receives either this binary SEM-derived mask or a procedurally generated binary filament mask and separates the dense network into individual filament instances.

FilaSeg is implemented in Python using NumPy, SciPy, scikit-image, OpenCV, Matplotlib, tifffile, and PyYAML. Its four processing stages are implemented as separate modules and coordinated by one end-to-end runner. The upstream CycleGAN/U-Net segmentation workflow is maintained separately from the FilaSeg implementation.

2.1 Workflow overview

The complete workflow contains two connected pipelines (Figure 1), described in turn below.

2.1.1 Binary segmentation of SEM images

The first pipeline converts a real SEM image into a binary mask representing the apparent axes of the CNT bundles.

Procedurally generated binary filament masks provide the structural mask domain, while real SEM images provide the microscopy-image domain. CycleGAN image translation is used to generate SEM-like images from the procedural masks. Because each generated SEM-like image originates from a known binary mask, the image and its source mask form a training pair for a U-Net segmentation model.

After training, the U-Net receives a real SEM image and predicts a binary CNT bundle-axis mask. This binary mask is then supplied to the FilaSeg instance-segmentation pipeline. The implementation of this upstream workflow is maintained separately [34].

This sequence is shown as Pipeline A in Figure 1.

The output is a thin binary mask representing the estimated bundle axes rather than the complete bundle silhouettes. Instance identity is not yet available at this stage: crossings appear as shared foreground regions, and several physical bundles may belong to one connected binary structure.

2.1.2 Instance segmentation of dense filament networks

The second pipeline receives a binary image of a dense filament network and separates it into individual filament instances.

The input can originate from either of two sources:

- a procedurally generated binary filament-axis mask used for development, validation, and comparison; or
- a binary CNT bundle-axis mask predicted from a real SEM image by the upstream CycleGAN/U-Net pipeline.

FilaSeg processes the binary mask through four consecutive stages:

1. tiled residual Hough vectorization;
2. orientation-aware fragment cleaning;
3. geometry-based instance reconnection; and
4. SEM-assisted apparent-width estimation and rendering.

Stages 1–3 use the binary filament-axis mask to recover individual centreline instances; Stage 4 uses a paired grayscale image to estimate their apparent width and render the final output (Section 2.3).

For a real CNT image, the paired grayscale image is the original SEM micrograph. For the present synthetic instance-segmentation experiments, the paired image is a fully procedural SEM-like rendering generated from the same known filament geometry.

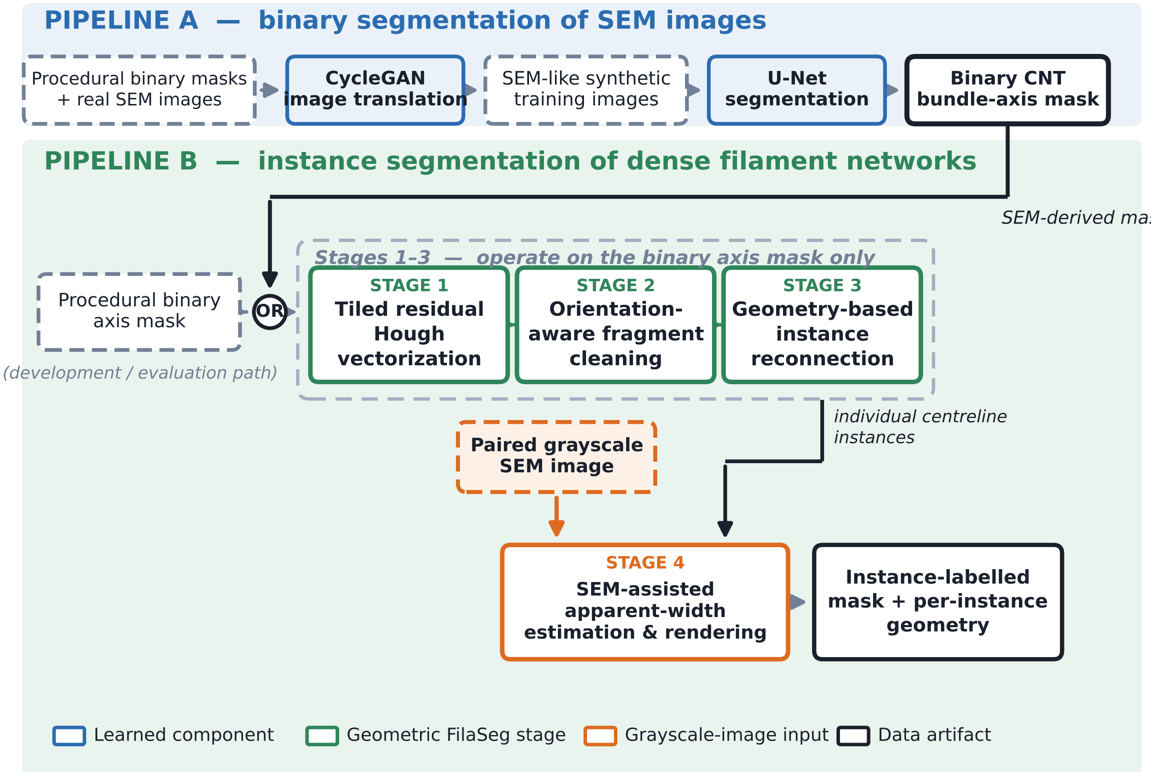


Figure 1. Relationship between the two pipelines. Pipeline A converts real SEM images into binary CNT bundle-axis masks. Pipeline B separates a dense filament network into instances from either this mask or a procedurally generated one. Stages 1–3 use the binary axis mask alone; Stage 4 also requires the paired grayscale SEM image.

Note: We currently evaluate the FilaSeg instance-segmentation pipeline separately from the upstream CycleGAN/U-Net pipeline. To discuss: should the final paper keep this separation, or present the two pipelines as one complete SEM-to-instance workflow?

2.2 Input images, masks, and reference annotations

The two pipelines use related synthetic data. For the simplified description in this manuscript, the procedural filament generator is treated as the common source of masks for both CycleGAN/U-Net development and FilaSeg evaluation.

The generator preserves the identity and geometry of every synthetic filament. It can therefore provide both binary foreground representations and exact instance-level references.

2.2.1 Procedural synthetic masks and SEM-like images

Each synthetic scene contains a set of filament instances

$$\mathcal{B} = \{B_1, \ldots, B_K\}, \tag{1}$$

where K is the number of generated filaments and B_k denotes the kth instance. Each filament is defined by an ordered centreline $\mathcal{C}_k$ and a thick rendered support Ω_k.

The individual supports are retained separately. Consequently, multiple filaments may own the same image location at a crossing. This produces an overlap-aware instance reference rather than a conventional label image in which each pixel can belong to only one object.

The following representations are derived from the same procedural geometry, as shown in Figure 2:

- an overlap-aware thick-instance reference;
- a combined binary foreground mask;
- an undegraded one-pixel filament-axis mask;
- a degraded thin-axis mask representing the expected output of an upstream binary segmenter; and
- one or more SEM-like grayscale representations.

The degraded axis masks may contain finite axis thickness, small gaps, irregularities, and other segmentation-like imperfections. These degraded masks are the direct inputs to the primary FilaSeg evaluation.

The nominal axis width names the generator's *stamping* width, not the width of the realized binary mask. Each filament axis is first rendered as a continuous-valued ridge of nominal width 1 px with an anti-aliased shoulder (soft-edge extent 0.6 px), then binarized at a low threshold (0.27) chosen to resemble a U-Net-like segmenter. Because a low threshold on a soft-shouldered ridge retains most of that shoulder, the realized degraded one-pixel mask measures approximately 1.7–1.8 px in mean foreground width. The undegraded one-pixel mask is instead drawn as a raw polyline with no ridge and no thresholding, so it measures almost exactly one pixel.

The synthetic scenes span five target areal-density levels from 20% to 60%. Areal density is calculated from the union of the thick filament supports, not from the thin centreline mask.

Two further scene descriptors are reported alongside areal density where the difficulty of the grouping problem is at issue. The *centreline-length density* is the total visible centreline arc length divided by the image area, in px px^{-2}; unlike areal density it does not depend on how wide the bundles are rendered. The *crossing density* counts pairwise centreline crossing events per megapixel of image area: for each pair of filaments, every connected region of pixels shared by their one-pixel centrelines counts as one event, so a junction where three centrelines meet contributes three pairwise events. Both quantities are measured on the clean centreline rasters and are therefore independent of the rendered bundle width and of the mask degradation.

SEM-like image generation. SEM-like synthetic images can be generated in two ways.

First, CycleGAN can translate procedural binary masks into images whose appearance resembles the real SEM domain. The source procedural masks remain available as paired labels for U-Net training.

Second, a fully procedural renderer can generate an SEM-like grayscale image directly from the known synthetic filament geometry. This approach does not require image translation and provides exact correspondence between the grayscale image, centreline mask, and instance reference.

The current FilaSeg synthetic experiments use the fully procedural SEM-like image as the paired grayscale input for the Stage 4 width and rendering analysis. The grouping evaluation itself is performed on the binary axis mask before Stage 4.

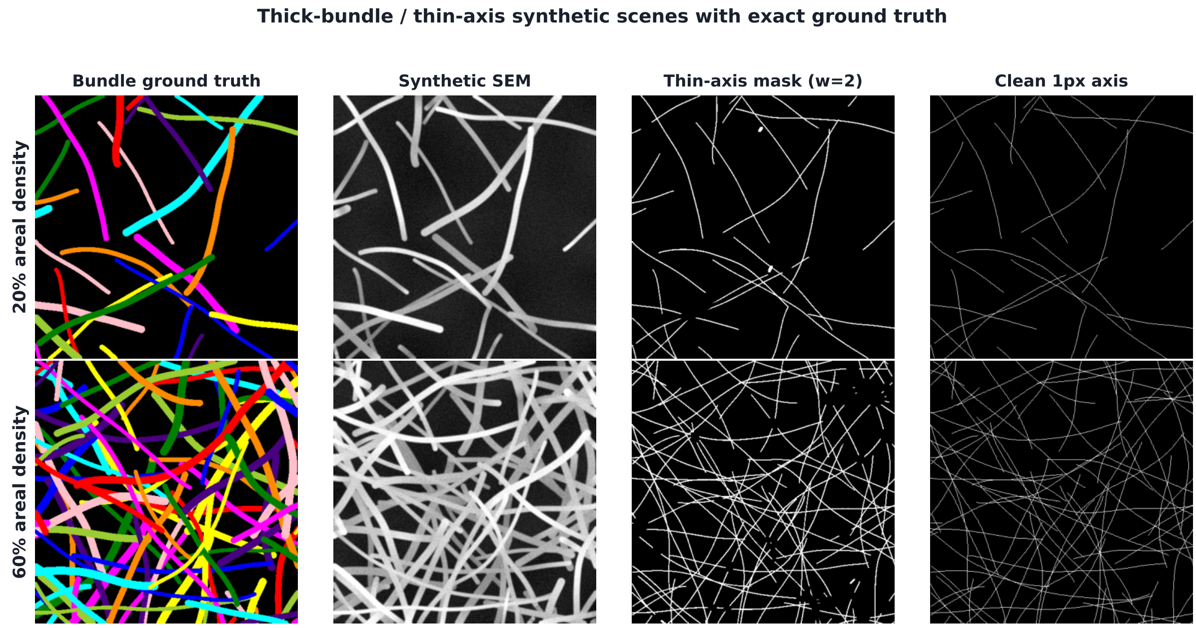


Figure 2. Examples of the procedural synthetic data (columns as labelled). The degraded thin-axis mask is shown at a nominal two-pixel width to make the degradation visible; the primary evaluation uses the nominal one-pixel variant. Instance identity is retained across all representations.

Note: The synthetic FilaSeg experiments currently use fully procedural SEM-like images during Stage 4. We intend to replace these with CycleGAN-generated SEM-like images so the synthetic Stage-4 input more closely matches the upstream segmentation workflow – evaluate the effect of that change before folding it into the primary results.

The same procedural mask generator is currently assumed to support both the CycleGAN/U-Net workflow and the FilaSeg evaluation. The final manuscript should document whether the generator settings, image dimensions, filament density settings (both areal and centreline-length), width distributions, and degradation settings are exactly the same in both uses.

2.2.2 Real SEM images and manual reference annotation

The real images used in this study are 8-bit grayscale secondary-electron SEM micrographs of dense CNT bundle sheets. The grayscale image is denoted by

$$S \in [0, 255]^{H \times W}, \tag{2}$$

where H and W are the image height and width in pixels.

The reference acquisition has a horizontal field width of 1.66 µm. The available metadata associates this field width with 1536 px, giving the reference physical pixel size

$$u_{\text{ref}} = \frac{1.66\ \text{µm}}{1536\ \text{px}} \approx 1.081 \times 10^{-3}\ \text{µm}\,\text{px}^{-1} \approx 1.08\ \text{nm}\,\text{px}^{-1}, \tag{3}$$

where u_{ref} is the physical size represented by one pixel in the reference image.

After removal of the instrument data bar, the stored micrographs measure 1534×1026 px. Intensity-dependent Stage 4 measurements are evaluated on the original 8-bit grayscale values without additional intensity normalization.

One 512×512 crop contains a manual reference annotation of 58 filament instances. This crop was used during method development and is therefore not a held-out test. It is included as a qualitative and development-stage example of application to a real CNT bundle network.

The corresponding binary bundle-axis mask was predicted by the upstream CycleGAN/U-Net workflow. The available project records do not contain a complete training-image list, data-split record, or training log for that upstream model. It cannot therefore be confirmed whether the crop, its parent image, or closely related images contributed to development of the binary segmenter.

The real annotation is treated as one human interpretation of an ambiguous dense image rather than as exact ground truth. Exact instance references are available only for the procedurally generated scenes.

[Needed before submission: Add the SEM instrument model, detector type, accelerating voltage, working distance, specimen preparation and CNT material identity before submission, and confirm the relation between the reported 1536-pixel calibration width and the stored 1534-pixel image width.]

2.3 FilaSeg instance-segmentation pipeline

The core FilaSeg input is a binary filament-axis mask

$$M \in \{0,1\}^{H \times W}, \tag{4}$$

where H and W are the image height and width. Each pixel of M is either zero for background or one for the detected filament axis. The mask therefore shows where filament axes are present, but it does not yet identify the individual filament instances.

At crossings, the mask contains one shared foreground region and does not identify which branch belongs to which physical filament. Small gaps may also interrupt otherwise continuous trajectories. FilaSeg must therefore infer how the visible fragments should be grouped into individual instances.

Stages 1–3 return a set of reconstructed centreline instances

$$\mathcal{I} = \{I_1, \ldots, I_{\widehat{K}}\}, \tag{5}$$

where $\widehat{K}$ is the number of predicted instances. Each instance contains an ordered centreline, endpoints, and derived geometric measurements. Stage 4 additionally uses a paired grayscale image to estimate apparent projected widths and render the reconstructed instances.

Physical-scale adaptation. FilaSeg's pixel-valued parameters were tuned using images represented at the reference physical pixel size u_{ref}. Examples include maximum gap lengths, minimum fragment areas, morphological kernel sizes, and curvature thresholds. If an image is acquired at a different magnification or pixel size, applying the same numerical pixel values would correspond to different physical distances and areas. The parameters are therefore rescaled so that they retain approximately the same physical meaning across images.

For a current image with physical pixel size u, the pixel-scale conversion factor is defined as

$$s_{\mathrm{px}} = \frac{u_{\mathrm{ref}}}{u}, \tag{6}$$

where u_{ref} is the physical size represented by one pixel in the reference image and u is the physical size represented by one pixel in the current image.

Finer sampling than the reference ($u < u_{\text{ref}}$) gives $s_{\text{px}} > 1$, so the same physical distance occupies more pixels; coarser sampling gives $s_{\text{px}} < 1$ and fewer pixels.

Parameters are scaled according to their physical dimension:

$$\begin{aligned} d_{\text{px}}(u) &= s_{\text{px}}\, d_{\text{px,ref}}, \\ A_{\text{px}}(u) &= s_{\text{px}}^2\, A_{\text{px,ref}}, \\ \kappa_{\text{px}}(u) &= \frac{\kappa_{\text{px,ref}}}{s_{\text{px}}}, \end{aligned} \tag{7}$$

where $d_{\text{px,ref}}$ is a reference distance or length measured in pixels, $A_{\text{px,ref}}$ is a reference area measured in pixels squared, and $\kappa_{\text{px,ref}}$ is a reference curvature threshold measured in inverse pixels. The corresponding quantities $d_{\text{px}}(u)$, $A_{\text{px}}(u)$, and $\kappa_{\text{px}}(u)$ are the values used for the current image.

Length-valued parameters scale linearly because a physical distance is represented by a different number of pixels. Area-valued parameters scale with s_{px}^2 because both image dimensions change. Curvature scales inversely because it has units of inverse pixels.

For example, consider an image whose pixel size is twice as large as the reference pixel size, so that $u = 2u_{\text{ref}}$ and $s_{\text{px}} = 0.5$. A reference maximum gap of 20 px is then reduced to 10 px, while a reference minimum area of 100 px^2 is reduced to 25 px^2. Both adjusted values represent approximately the same physical distance and area as in the reference image.

Dimensionless parameters, such as fractions, angular thresholds expressed as cosines, and iteration counts, are not rescaled. The physical pixel size of the current image is supplied explicitly or obtained from available SEM acquisition metadata. The reference scale is used only as a documented fallback when image-specific scale information is unavailable.

2.3.1 Stage 1: tiled residual Hough vectorization

Stage 1 converts the binary axis mask into a set of short, oriented line primitives. It uses a tiled, orientation-binned residual decomposition based on the progressive probabilistic Hough transform [35].

Within each tile and orientation bin, line segments are selected to explain currently unassigned foreground pixels. This produces local orientation estimates and may bridge small raster interruptions along an otherwise continuous filament trajectory.

The tiled design allows large images to be processed locally, while overlap between neighbouring tiles reduces boundary artefacts. Selected vectorization parameters also adapt to the observed foreground width.

[To do: Add the tile dimensions, orientation bins, overlap rules, residual-update logic and stopping criteria.]

2.3.2 Stage 2: orientation-aware fragment cleaning

Stage 2 cleans the orientation-associated fragments produced by Stage 1. Small artefacts are removed, short along-filament gaps are closed using orientation-matched morphology, and each retained fragment is reduced to a smooth curve with two well-defined tips.

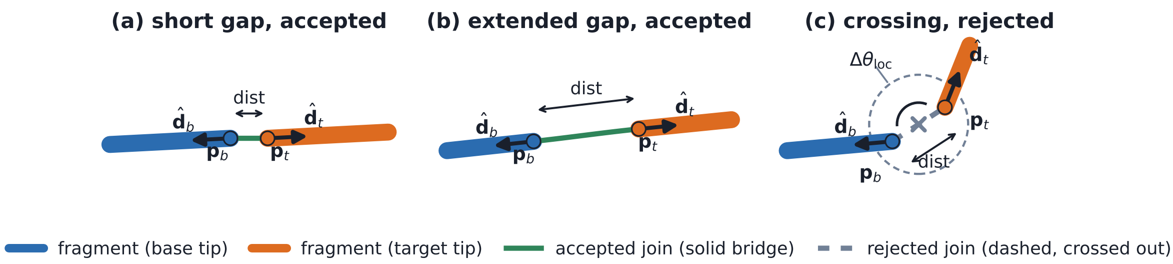


Figure 3. Representative reconnection decisions (see panel titles and legend for the accepted/rejected criteria in each case). Numerical thresholds are to be added.

The purpose of this stage is to provide the reconnection stage with physically plausible fragments whose endpoint positions and local directions are not dominated by raster noise, isolated pixels, or small branches.

[To do: Add the morphology settings, skeleton-cleaning operations, smoothing rules and branch-removal criteria.]

2.3.3 Stage 3: geometry-based instance reconnection

Stage 3 merges compatible fragments into individual filament instances. Each fragment is represented by its centreline, endpoints, local endpoint directions, and associated geometric descriptors.

Candidate endpoint pairs must satisfy a sequence of geometric criteria involving endpoint separation, orientation consistency, collinearity, bridged-path smoothness, and local topology. An orientation-based crossing-prevention criterion reduces the risk of joining fragments that belong to different filaments but meet near a crossing.

Because filament axes are unoriented, the difference between two axis angles θ_1 and θ_2 is defined as

$$\Delta\theta = \min(|\theta_1 - \theta_2|, 180^\circ - |\theta_1 - \theta_2|) \in [0^\circ, 90^\circ], \tag{8}$$

where $\Delta\theta$ is the smallest angular difference between the two unoriented axes.

Candidate merges are evaluated in ordered passes. The earliest pass handles short, high-confidence gaps. Later passes consider geometrically more difficult cases. A topology-checked repair procedure addresses configurations that cannot be recovered through direct endpoint-to-endpoint continuation. Figure 3 shows representative examples of accepted and rejected candidate merges.

[To do: Add the candidate scoring, pass-specific thresholds, merge ordering, crossing-prevention rules, topology tests and repair logic.]

2.3.4 Stage 4: SEM-assisted apparent-width estimation and rendering

Stage 4 combines the reconstructed centreline instances with a paired grayscale image. For each instance, transverse intensity profiles are sampled at positions along its centreline. Opposite-polarity intensity changes on either side of the centreline are used to estimate one apparent projected width for the instance.

The measured quantity is described as an *apparent projected width* rather than a diameter. The SEM image provides a two-dimensional projected intensity profile, and the method does not assume that the physical bundle cross-section is circular.

Each centreline is rendered at its estimated width. Conflicts introduced by width rendering are handled while retaining multi-label ownership at genuine crossings, where more than one physical bundle may occupy the same projected image location.

Stage 4 is not part of the primary centreline-grouping endpoint. Its apparent widths and rendered outputs require validation separately from the Stage 3 instance-grouping result.

Note: To discuss: should Stage 4 stay part of the main FilaSeg pipeline, or be presented as a separate SEM-assisted measurement and postprocessing module?

2.4 Evaluation metrics, comparison protocol, and statistical analysis

FilaSeg is evaluated primarily as a binary-mask-to-instance method. The procedural evaluation scenes provide exact instance references, while the single real SEM crop is treated only as a development case study.

Synthetic development scenes and the manually annotated real crop were used to develop the method and select comparison-method settings. The FilaSeg configuration, comparison settings, and metric implementation were then fixed before evaluating a disjoint set of previously unevaluated synthetic scenes.

The predefined synthetic evaluation contains 125 independently seeded scenes distributed across five target areal-density levels. Each comparison method receives the same degraded binary mask for a given scene.

The development set contains 20 procedural geometries together with the single real crop. The disjoint predefined set was generated from base seed 20260801 with 25 scenes at each target areal density: for areal density $d \in \{20, 30, 40, 50, 60\}$ and zero-based index i, the scene seed is $20260801 + 100d + i$. Exact scene paths, seeds, input and ground-truth hashes, method outputs and failure states are recorded per scene in the evaluation manifest distributed with the manuscript.

2.4.1 Comparison methods

The principal same-input comparator is a minimum-turn skeleton tracer. It skeletonizes the binary mask and continues branches through junctions or gaps by selecting the continuation with the smallest turning angle, subject to a maximum allowed gap and turning angle selected during development.

A connected-component method is retained only as an additional lower-bound sanity check. It assigns one label to each connected foreground component and does not attempt to resolve crossings or reconnect gaps.

Component-ablation experiments remove selected FilaSeg criteria while retaining the remainder of the pipeline. These experiments are exploratory and are used to investigate which processing stages contribute to the observed performance.

2.4.2 Primary metric: common-fragment pairwise F_1

The primary metric is a fragment-level adaptation of pairwise clustering precision, recall, and F_1. Pairwise comparison of partitions originates from the Rand family of clustering measures [36] and was later adapted to quantify split and merge errors in image and volume segmentation [37].

For each synthetic scene, FilaSeg and the minimum-turn tracer receive the same degraded binary filament-axis mask. Each method independently separates and groups the visible filament structure into predicted instances.

For scoring, the visible input structure is partitioned into a common, method-independent set of elementary centreline fragments. These fragments are not additional inputs to the algorithms. They serve only as shared scoring units. Each fragment is assigned a reference filament identity from the procedural ground truth and a predicted instance identity from each method's output.

This fragment-level adaptation is appropriate because the main objective is to determine which visible parts of the network belong to the same physical filament. It also prevents arbitrary instance numbering, centreline thickness, and Stage 4 width rendering from influencing the grouping score.

For all valid pairs of common fragments:

- a true-positive pair contains two fragments from the same reference filament that are assigned to the same predicted instance;
- a false-positive pair contains fragments from different reference filaments that are incorrectly assigned to the same predicted instance; and
- a false-negative pair contains fragments from the same reference filament that remain assigned to different predicted instances.

Pairwise precision, recall, and F_1 are calculated as

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}, \qquad R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}, \qquad F_1 = \frac{2PR}{P + R}. \tag{9}$$

Pairwise precision decreases when fragments from different filaments are over-merged into one predicted instance. Pairwise recall decreases when fragments from one filament remain split across several predicted instances.

Before fragment identities are assigned for scoring, every method output is reduced to an equivalent one-pixel centreline representation. The metric therefore evaluates instance grouping independently of Stage 4 apparent-width estimation and rendering.

Fragments that cannot be assigned to a valid reference filament are retained as singleton false evidence rather than being silently excluded from the evaluation.

In addition to the pairwise grouping score, an object-level recovery summary is reported. Each reference filament is classified according to whether it is recovered as one dominant predicted instance, fragmented across several predicted instances, or absorbed into another instance. Predicted instances without a dominant reference association are counted as false instances. This companion analysis helps interpret whether a reduction in pairwise precision or recall arises mainly from over-merging, fragmentation, or loss of complete filament instances.

2.4.3 Statistical analysis

Uncertainty is estimated using paired, density-stratified bootstrap resampling of the synthetic evaluation scenes. Pairing preserves the comparison between methods within each scene, while density stratification preserves the number of scenes contributed by each target areal-density level.

The primary reported contrast is the within-scene difference between FilaSeg and the minimum-turn comparator. Confidence intervals are calculated from the bootstrap distribution of the corresponding aggregate statistic.

Density-specific results are reported separately because the difficulty of the grouping problem changes as the synthetic filament network becomes denser.

[To do: Add the metric edge cases, object-recovery thresholds, bootstrap replicate counts, random seeds and complete ablation configurations.]

3 Results

The primary quantitative comparison uses the method-independent centreline fragments and equivalent one-pixel instance representations defined in Section 2.4.2. Confidence intervals for overall results use density-stratified scene resampling, while differences between methods are paired within each scene. Stage 4 width rendering and the connected-component lower-bound check are treated as secondary analyses.

3.1 Validation of FilaSeg instance grouping

FilaSeg was compared with the minimum-turn skeleton tracer on 125 independently seeded synthetic scenes. Both methods received the same degraded binary filament-axis mask for each scene, and their outputs were evaluated using the same elementary centreline fragments.

Quantitative comparison. FilaSeg Stage 3 achieved a mean common-fragment F_1=0.734 with a density-stratified 95% bootstrap confidence interval of [0.722, 0.745]. Its mean pairwise precision and recall were 0.749 and 0.724, respectively.

Table 1 and Figure 4 summarize the comparison with the minimum-turn tracer.

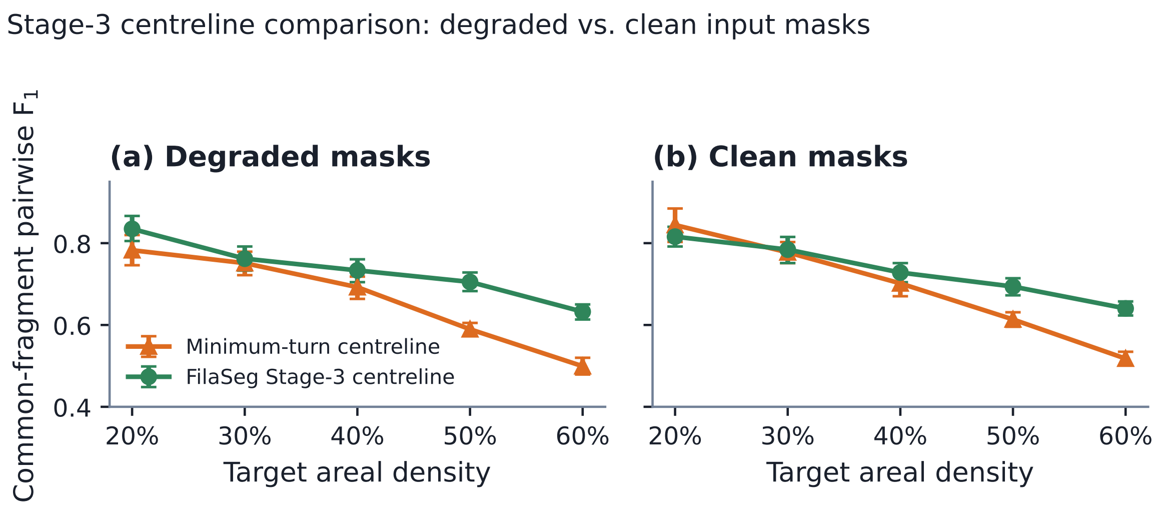
Table 1. Predefined synthetic comparison of FilaSeg Stage 3 and the minimum-turn tracer, scored on equivalent one-pixel centreline representations. F_1 intervals and the paired difference use density-stratified bootstrap resampling; object-level recovery is the companion measure defined in Section 2.4.2.

Method	P	R	F_1	Object-level recovery
Minimum-turn skeleton	0.725 [0.712, 0.738]	0.616 [0.602, 0.631]	0.663 [0.651, 0.675]	0.513 [0.497, 0.528]
FilaSeg (Stage 3 centreline)	0.749 [0.735, 0.763]	0.724 [0.710, 0.737]	0.734 [0.722, 0.745]	0.591 [0.575, 0.607]
Paired difference (FilaSeg − minimum-turn)	0.024 [0.009, 0.038]	0.107 [0.094, 0.120]	0.071 [0.059, 0.082]	0.079 [0.062, 0.096]

The paired FilaSeg-minus-minimum-turn difference was 0.071, with a 95% confidence interval of [0.059, 0.082].

The aggregate interval excludes zero, supporting an advantage for FilaSeg over the selected minimum-turn comparator under the tested generator and evaluation conditions. The advantage was not uniform across density. The density-specific interval crossed zero at the 30% areal-density level (0.011 ± 0.073; 95% CI [-0.017, 0.039] (n=25)). The result therefore supports a conditional advantage over this comparator rather than a general superiority claim.

The connected-component lower bound reached a mean common-fragment F_1 of only 0.077, confirming that the grouping problem is not solved by treating each connected foreground region as one instance.



Same 125 scenes in both panels (25 per density level), paired scene-by-scene. Markers show the per-density mean common-fragment F_1 (individual scenes not plotted); error bars are 95% percentile bootstrap CIs over scenes (seed 20260729, 10000 resamples). (a) mask_w1.png -- the primary, predefined input (degraded/UNet-realistic; mean foreground width ~1.8 px, area/skeleton-length). (b) mask_clean.png -- a secondary, non-predefined analysis on the undegraded, exactly-1px axis mask.

Figure 4. Common-fragment F_1 for FilaSeg Stage 3 and the minimum-turn tracer, by target areal-density level: (a) predefined degraded mask; (b) secondary, non-predefined analysis on the same 125 scenes with the undegraded one-pixel mask. Per-density means; 95% bootstrap CIs (25 scenes/level).

Where the advantage comes from: degraded versus clean input. The predefined comparison supplies both methods a degraded axis mask, so its paired difference mixes two abilities: grouping already-visible structure, and recovering structure that the modeled segmentation damage has interrupted. To separate them, the same 125 scenes were re-scored with the undegraded, exactly one-pixel axis mask in place of the degraded input, holding the configuration and the metric fixed. This is a secondary analysis: the scenes are the predefined ones, but the input mask differs and the comparison was not specified before the configuration was frozen.

The two methods respond very differently (Figure 4b). The minimum-turn tracer improved when given clean input, from 0.663 to 0.691, a paired change of +0.028 with a 95% confidence interval of $[0.017, 0.040]$ that excludes zero. FilaSeg did not change: 0.733 against 0.734, a paired change of -0.001 with an interval of $[-0.012, 0.010]$ spanning zero. The FilaSeg-minus-minimum-turn margin therefore narrows from 0.071 [0.059, 0.082] on degraded input to +0.042 $[0.031, 0.053]$ on clean input.

The FilaSeg-minus-minimum-turn margin on clean input remains resolved and is largest where crossings rather than gaps dominate: at 60% areal density FilaSeg reached 0.640 against 0.518 for the tracer even with clean masks.

Stage-3 and Stage-4 representation analysis. Because the grouping score depends on which representation is submitted for scoring, the same FilaSeg output was evaluated at three points in the pipeline. Reducing the Stage 4 output back to a centreline representation changed the common-fragment F_1 by only -0.009 relative to the Stage 3 centrelines (95% CI [-0.011, -0.006]). Scoring the width-rendered Stage 4 masks instead changed it by -0.053 (95% CI [-0.059, -0.047]).

The larger part of that change therefore arises from width rendering rather than from any change of grouping decision (-0.044 between the rendered and centreline Stage 4 representations). Equivalent Stage-3 centrelines are used as the primary grouping representation throughout, and the width-rendered output is reported only as a secondary diagnostic.

Largest FilaSeg wins

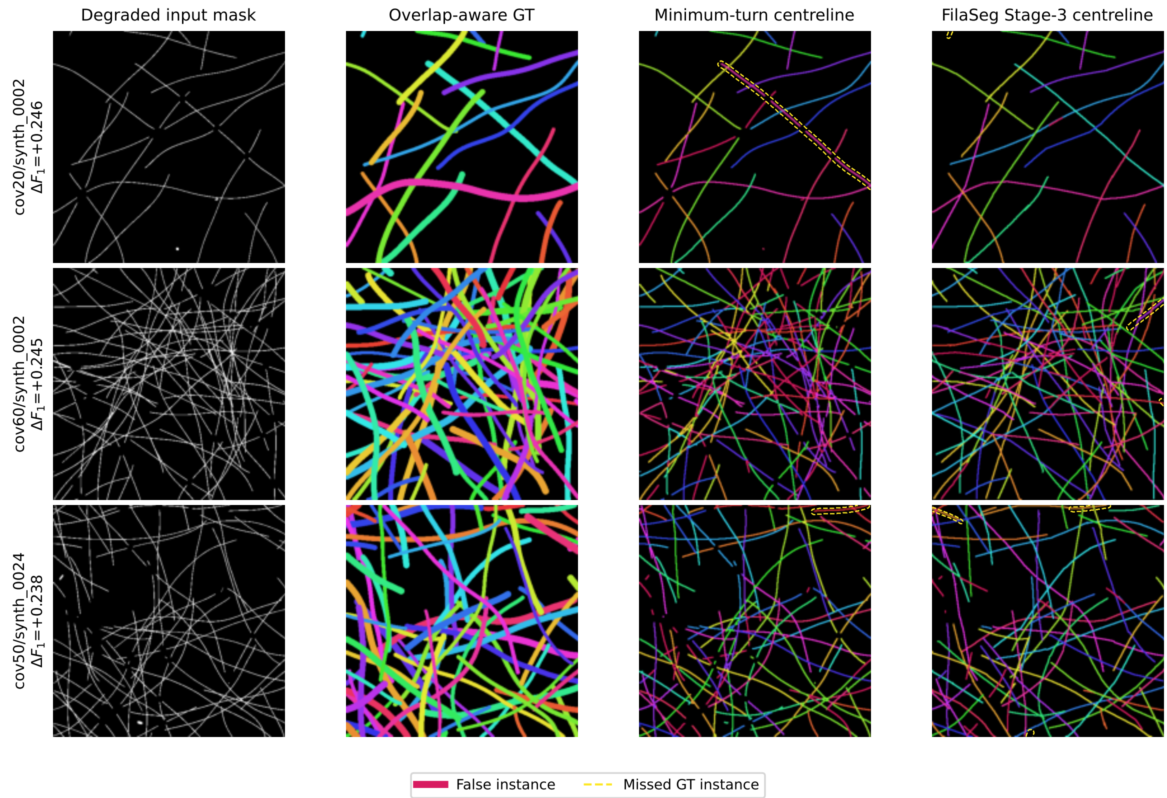


Figure 5. The three predefined scenes with FilaSeg's largest paired advantage under the primary common-fragment endpoint, selected from stored numerical differences rather than visual inspection (Section 3.1).

Qualitative wins and losses. Figures 5 and 6 show the three scenes with the largest paired advantage for each method. Scene selection used the stored numerical differences alone, not visual inspection, so both directions of error are represented; predicted and reference centrelines in both figures use a 3-pixel display-only dilation for print legibility, and the underlying one-pixel skeletons used for matching and scoring are unchanged. The loss cases in Figure 6 are genuine counterexamples: they show conditions under which local minimum-turn continuation grouped the visible filament fragments more accurately than FilaSeg.

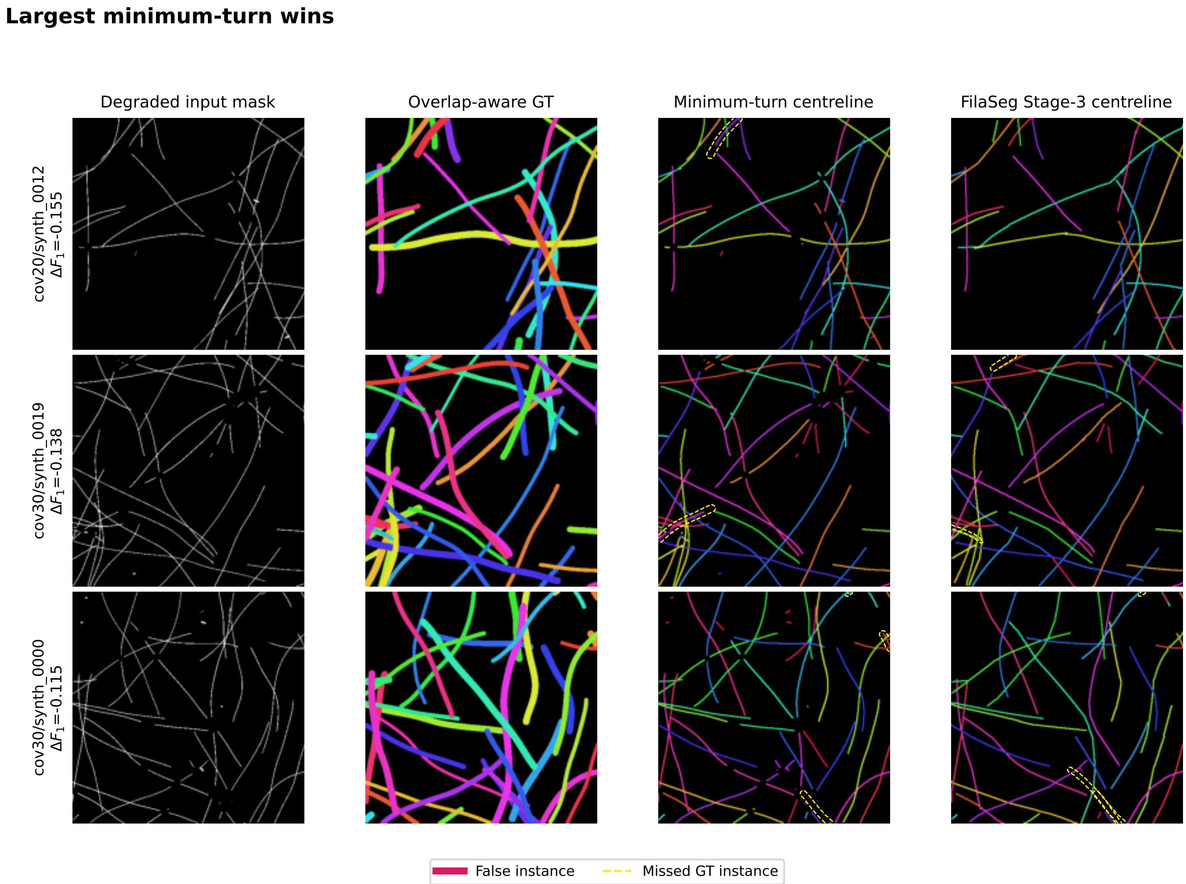


Figure 6. The three predefined scenes with the minimum-turn tracer's largest paired advantage, selected the same way as Figure 5; these are genuine counterexamples (Section 3.1).

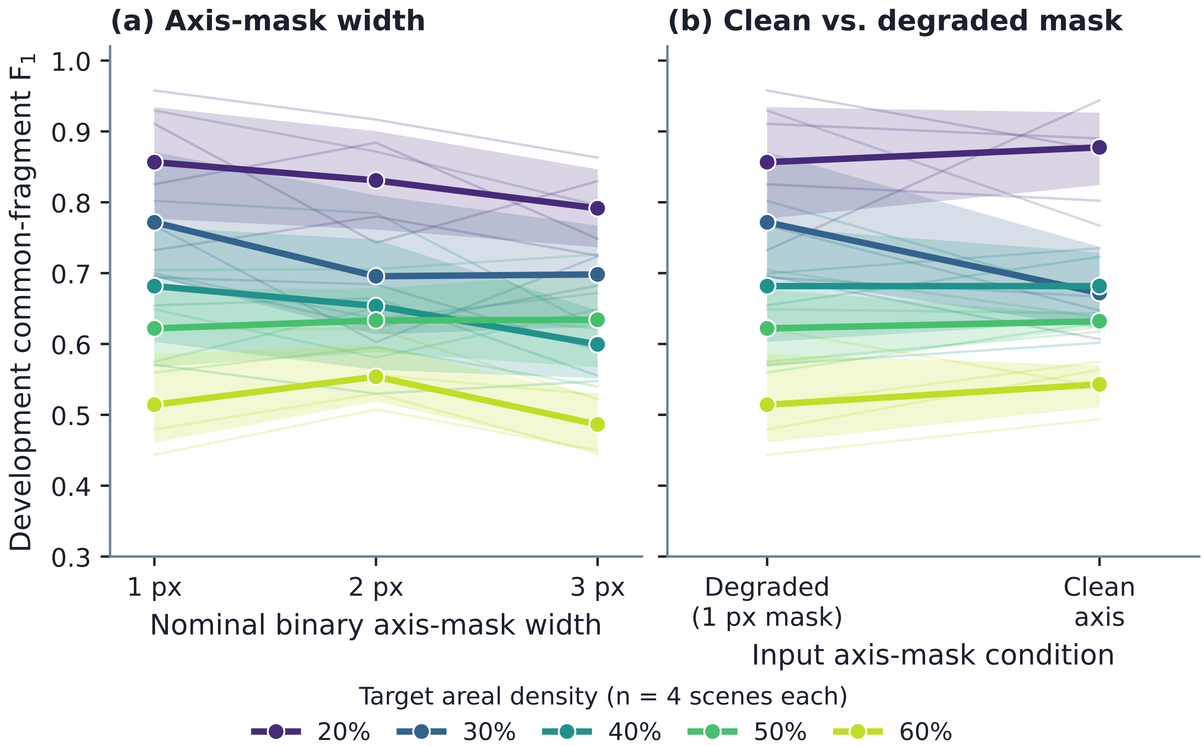


Figure 7. Sensitivity of the common-fragment F_1 to the input representation, by target areal density (Section 3.2); panels as labelled. Curves are means over 4 scenes per level (20 total); bands are 95% bootstrap CIs preserving within-scene pairing (10 000 replicates).

3.2 Sensitivity to input-mask and synthetic scene properties

These experiments use development data and are therefore interpreted as sensitivity analyses rather than independent validation. Because the same scenes recur at every condition, their confidence intervals come from a percentile bootstrap that resamples scenes within an areal-density level and reuses the resampled scene set across conditions, preserving the within-scene pairing.

Axis-mask width and input degradation. The effect of the binary filament-axis width was evaluated on a 20-scene development set using masks with nominal widths of one, two, and three pixels (Figure 7a). The one-pixel masks produced the highest mean common-fragment F_1. The paired confidence interval for the one-pixel-versus-two-pixel difference included zero (unresolved at the 95% level). The three-pixel masks produced a larger decrease whose paired interval excluded zero. A thin axis representation is therefore preferable under the tested configuration, particularly once the foreground becomes several pixels wide.

The degraded one-pixel masks were also compared with the corresponding clean procedural axes, which omit the modeled segmentation-like imperfections such as small gaps and local raster irregularities (Figure 7b). Replacing the degraded masks with clean procedural axes produced a small negative mean change whose paired confidence interval included zero. The analysis therefore did not show a resolved aggregate improvement from removing the modeled degradation. At 30% areal density all four available development scenes scored lower for the clean representation, but four scenes cannot support a density-specific conclusion. This is a conditional development diagnostic, not evidence that segmentation errors are unimportant.

Separating centreline-length density from bundle width. The target areal-density levels used above are computed from the union of the thick rendered filament supports. Areal density therefore rises both when the network contains more or longer centrelines and when the rendered bundles are wider, so it cannot by itself attribute difficulty to either cause. A separate development generator was used to break that coupling: it stops on summed visible centreline arc length rather than on areal density, which allows the target bundle width to be varied at fixed centreline geometry.

The resulting development-only factorial spans three centreline length-density levels (0.02, 0.04 and 0.06 px px^{-2}) and three target bundle-width levels (6, 11 and 16 px), with four independently seeded scenes per cell and 36 scenes in total. Seeds are disjoint from the predefined evaluation set.

The design yields an exact structural control. Within each seed and length-density block, the degraded one-pixel input mask, the clean centreline mask and the overlap-aware centreline reference are byte-identical across all three width levels, verified by SHA-256 comparison, while achieved thick-mask areal density varies by more than a factor of two across those same cells. Crossing density, gap count, filament count and one-pixel axis coverage are likewise invariant across width within a block. Because Stages 1–3 consume only the binary axis mask, target bundle width is therefore an exact negative control for centreline grouping, and thick-mask areal density cannot by itself change the grouping result. Width acts as an active factor only for the SEM-like rendering and the Stage 4 apparent-width measurement.

The empirical arm confirms the structural argument. Scoring the Stage-3 centreline output of all 36 scenes with the same common-fragment metric used for the primary evaluation, the maximum within-geometry range of common-fragment F_1 across the three width levels was exactly zero in all twelve seed-by-density blocks, even though achieved thick-mask areal density varied by a factor of two to two-and-a-half across those same cells. Rendered bundle width, and the areal density it produces, therefore have no effect whatsoever on centreline grouping. Because the three width levels are exact replicates of one grouping problem rather than independent conditions, they are pooled in what follows, and width is not shown as a separate series.

Grouping performance instead fell monotonically with the geometric complexity of the centreline network. Mean common-fragment F_1 decreased from 0.859 at 118 crossings per megapixel to 0.803 at 498 and 0.713 at 1126, with pairwise precision falling from 0.875 to 0.693, pairwise recall from 0.846 to 0.736, and object-level recovery from 0.745 to 0.531 over the same range. Increasing centreline-length density raised crossing density and gap count far more steeply than it raised centreline length itself, identifying network geometric complexity, rather than areal density, as the quantity that makes grouping harder.

Density sweep against the minimum-turn tracer. Because bundle width is inert for Stages 1–3, the centreline-length-density axis was extended and both methods compared on it directly. Two further length-density levels (0.01 and 0.03 px px^{-2}) were generated with the same four seeds and the same generator settings, giving five levels spanning 23 to 1126 crossings per megapixel at a single target bundle width of 6 px. FilaSeg and the minimum-turn tracer received the identical degraded axis mask for every scene, so the two are paired scene by scene (Figure 8).

The two methods respond to centreline-length density very differently. At the sparsest level the minimum-turn tracer was the better of the two, reaching a mean common-fragment F_1 of 0.921 against 0.853 for FilaSeg. The ordering reverses as the network fills in: 0.859 against 0.812 at 0.02 px px^{-2}, 0.833 against 0.799 at 0.03, 0.803 against 0.714 at 0.04, and 0.713 against 0.560 at 0.06. FilaSeg loses roughly 0.14 F_1 across the whole range while the tracer loses roughly 0.36.

Table 2. Achieved properties and Stage-3 outcomes of the nine-cell development factorial (four seeds/cell, 36 scenes). Crossing density, gap count and all grouping scores are width-invariant within a geometry block; thick-mask areal density is the only quantity that responds to width.

Target length density	Achieved length density	Crossings per Mpx	Mean gap count	Thick-mask areal density by width			Common F_1	Pairwise P	Pairwise R	Object-level recovery
				6 px	11 px	16 px	(pooled over widths; width-invariant, n=4 seeds)			
0.020	0.0208	118.3	13.0	0.121	0.212	0.295	0.859	0.875	0.846	0.745
0.040	0.0407	497.8	49.2	0.222	0.372	0.494	0.803	0.808	0.801	0.707
0.060	0.0607	1126.3	97.5	0.310	0.496	0.635	0.713	0.693	0.736	0.531

Common-fragment F_1/P/R/object recovery: FilaSeg Stage-3 (pre-render) centreline output vs. overlap-aware centreline ground truth; exactly width-invariant within every (seed, density) block (max. range across widths = 0.0); pooled over widths, averaged over 4 seeds.

This is the mechanism behind the conditional advantage reported in Section 3.1. When filaments are sparse and crossings are rare, local minimum-turn continuation is not merely adequate but slightly better, because almost every junction is unambiguous and the additional geometric criteria can only reject continuations that were in fact correct. The advantage of the multi-criterion approach appears once crossings become common enough that the locally smoothest continuation is frequently the wrong one. With four seeds per level, the crossing of the two curves is consistent with the density-specific behaviour of the predefined evaluation, where the interval also failed to resolve at the lowest areal-density levels.

These values are not comparable with the primary evaluation in Section 3.1: this development generator stops on centreline arc length rather than areal density and therefore produces a different scene population. With four seeds per cell, no claim specific to centreline-length density is made from it. Its purpose is the attribution question, which it answers cleanly because the width control is exact rather than estimated. Table 2 reports the achieved scene properties and outcomes of all nine cells.

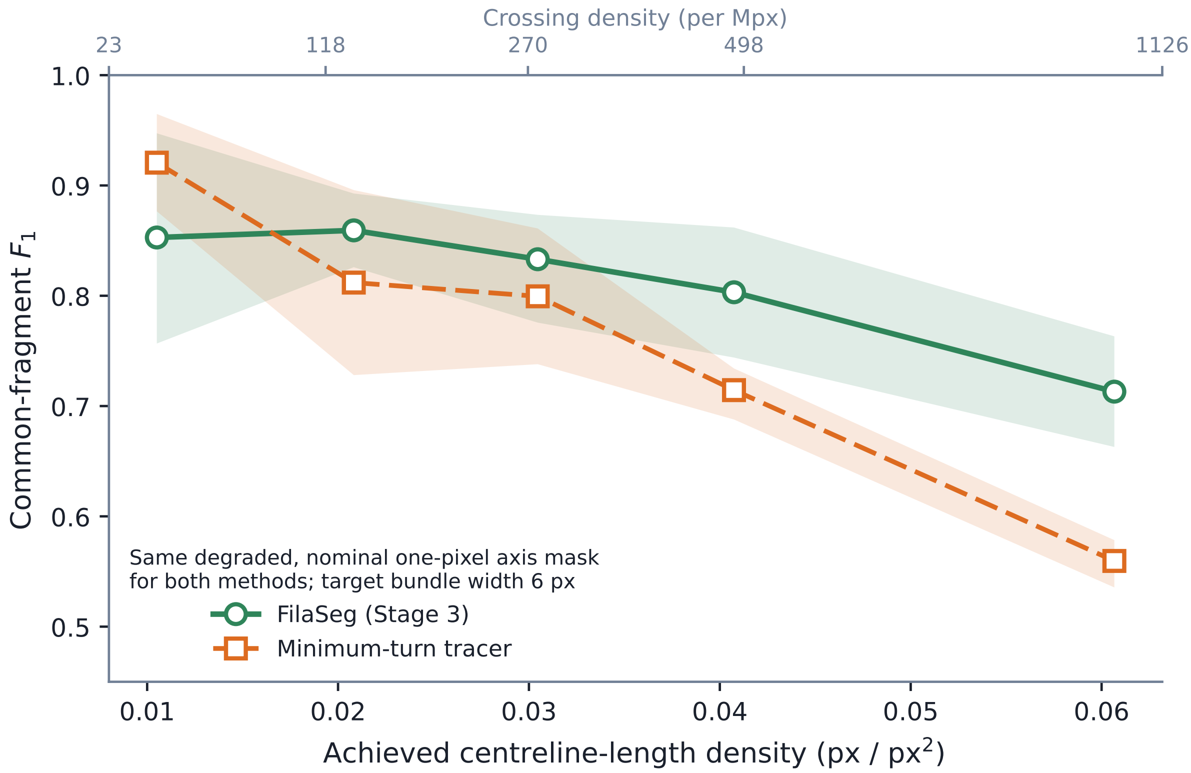


Figure 8. Density sweep (Section 3.2): common-fragment F_1 for FilaSeg Stage 3 and the minimum-turn tracer against achieved centreline-length density, paired scene by scene at the single inert target bundle width of 6 px. Bands are 95% bootstrap CIs of the mean ($n = 4$ seeds/density, 10 000 replicates).

3.3 Real SEM development case study

FilaSeg was also applied to the real SEM development crop described in Section 2.2.2 (a 512×512 crop with 58 manually annotated filament instances). As noted there, the crop informed method development; it does not provide an independent estimate of generalization or a population confidence interval, and all real-data observations in this manuscript derive from it.

Figure 9 shows the final rendered output. Predicted instances are assigned the colour of their one-to-one matched manual reference instance. Unmatched predicted instances are shown in magenta, and the unmatched manual reference instance is indicated by a dashed yellow outline.

The rendered instances are displayed because this figure illustrates the delivered Stage 4 output. It should not be interpreted as an independent validation of apparent-width accuracy or as the primary centreline-grouping endpoint.

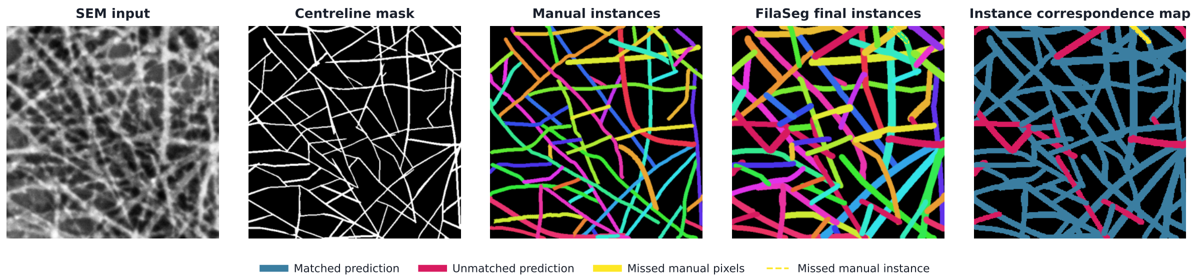


Figure 9. Real SEM development crop. Matched manual reference and predicted instances share colour after maximum-total-IoU assignment (unmatched/missed markers in legend). The final panel summarizes correspondence without treating apparent-width disagreement as a grouping error.

4 Discussion

Dense filament networks defeat the usual instance-segmentation assumption that each pixel belongs to one object. In a projected SEM image of a carbon-nanotube bundle sheet, filaments cross, merge visually, and are interrupted by segmentation failures, so the binary axis mask that an upstream segmenter produces contains no instance identity at all. The question addressed here is narrow and deliberately so: given that mask, which visible pieces of the network belong to the same physical filament?

On 125 independently seeded scenes of the primary synthetic evaluation, FilaSeg Stage 3 reached a mean common-fragment $F_1 = 0.734$ against 0.663 for the minimum-turn skeleton tracer receiving the identical mask, a paired difference of 0.071 with a 95% confidence interval of [0.059, 0.082] that excludes zero. The interval is an aggregate statement, not a uniform one: at 30% areal density the density-specific interval crosses zero, and the qualitative counterexamples in Section 3.1 contain scenes the minimum-turn tracer groups better.

First, the primary endpoint is Stage 3 centreline grouping, scored on one-pixel representations after every method output is reduced to an equivalent form (Section 2.4.2); Stage 4 apparent-width estimation and rendering are downstream of the endpoint and are not validated by it. Second, the evidence comes from one procedural generator and one fixed comparator. What follows therefore interprets a conditional advantage over a specific same-input alternative under specific synthetic conditions, and identifies what would be needed to say more.

4.1 General interpretation and scope

Why a geometric, non-learning method is useful here. Supervised instance segmentation of these images is limited by the reference data rather than by model capacity. Exact instance identity in a dense crossing network is not reliably annotatable by hand: the single real crop used here is one human interpretation of an ambiguous image, not ground truth. A method built from explicit geometric criteria avoids that dependency, needs no instance-labelled training set, and exposes every decision as an inspectable rule with a physical meaning. It also transfers across acquisition scales by construction, because its pixel-valued parameters are rescaled to hold their physical interpretation (Equations 6–7). The cost is that the rules are hand-specified, so the method cannot learn a regularity the designer did not anticipate.

Why FilaSeg improves on minimum-turn continuation on average. The minimum-turn tracer resolves each junction locally, choosing the smallest turning angle available at that point. This is exactly the decision that is least determined by the data in a dense network, because near a crossing several continuations have similar turning angles while belonging to different filaments. FilaSeg's gain comes from deferring and constraining that decision: candidate joins must satisfy endpoint separation, orientation consistency, collinearity and bridged-path smoothness together rather than turning angle alone, and they are considered in ordered passes so that short high-confidence gaps are resolved before geometrically ambiguous ones. The improvement is larger in pairwise recall than in precision, which is consistent with the mechanism: an evidence-ordered procedure with an explicit crossing-prevention criterion can accept long continuations that a purely local rule refuses, without accepting proportionally more wrong ones.

How much of the advantage is robustness to degradation, versus better grouping. Re-scoring the same 125 scenes on the undegraded, exactly one-pixel axis mask (Section 3.1) separates two contributions the paired degraded-input comparison mixes together: recovering structure that mask degradation interrupted, and grouping structure that was already fully visible. The minimum-turn tracer improved on clean input while FilaSeg did not, narrowing the margin from 0.071 to $+0.042$ $[0.031, 0.053]$. Much of FilaSeg's aggregate advantage over this comparator is therefore attributable to insensitivity to mask degradation rather than to better grouping of intact topology. The advantage does not vanish on clean input, and, as reported in Section 3.1, it remains largest at the highest tested crossing density even with clean masks.

Why the advantage is conditional rather than universal. The same ordering that recovers long filaments becomes a liability where the geometry genuinely under-determines the answer. At 30% areal density the paired interval includes zero, and the losses in Section 3.1 show the mechanism: where two filaments cross at a shallow angle and both are locally smooth, the multi-criterion test and the minimum-turn rule are choosing between hypotheses that the image does not distinguish. In that regime a global geometric argument has no advantage over a local one, and either can win by chance on a given scene. The aggregate interval should therefore be read as evidence that the multi-criterion approach helps across the tested density range on average, not that it dominates at every density.

Relationship to the upstream segmentation pipeline. FilaSeg consumes a binary bundle-axis mask, which in the real case study was produced by the CycleGAN/U-Net workflow summarized in Section 2.1.1 and Figure 1. The two pipelines are evaluated separately here, and deliberately so: supplying every method the identical mask isolates grouping from segmentation quality, which is the only way the comparison reported in Section 3.1 can be attributed to grouping at all. The corollary is that no end-to-end SEM-to-instance performance figure is established by these results. An end-to-end number would require propagating upstream segmentation error into the grouping stage on images the upstream model has demonstrably never seen.

What the sensitivity analyses do and do not show. The input-representation analyses in Section 3.2 are development-only diagnostics on small samples. They indicate that a thin axis representation is preferable, with the one-pixel and two-pixel comparison unresolved at the 95% level and a resolved decrease at three pixels; this is consistent with the vectorization stage estimating local orientation from a foreground whose thickness begins to compete with the geometry it is meant to measure. Removing the modeled degradation did not produce a resolved aggregate improvement – a limit of this generator's degradation model and its small per-density samples, not evidence that segmentation errors are unimportant in practice.

Separating network complexity from bundle width. Target areal density conflates two distinct things, because the union of thick filament supports grows both when the network contains more centreline and when the rendered bundles are wider. The factorial study (Section 3.2, Table 2) isolates them: bundle width is an exact structural negative control for Stage 3, and the common-fragment F_1 range across width levels was exactly zero in every geometry block, so thick-mask areal density cannot by itself drive grouping performance. The difficulty that rises with centreline-length density is instead the geometric complexity of the network – crossing density and gap count outpacing centreline length – not areal density; width remains an active factor only for SEM rendering and Stage 4, where it is the quantity being measured. This is an attribution result that the exact control settles cleanly; the four-seed cells remain too small to support centreline-length-density-specific effect sizes.

Applicability beyond CNT bundles. FilaSeg's criteria refer to filament geometry rather than to nanotube physics, so the approach is in principle applicable to other dense curvilinear networks imaged in projection, such as nanowire mats, fibre assemblies or cytoskeletal filaments. That is a direction for future work rather than a result: no data outside the CNT bundle setting were examined here, and each new domain changes the crossing statistics, the degradation model, and the scale metadata on which the parameter rescaling depends.

4.2 Failure modes and limitations

Residual grouping errors have two opposite directions. Errors are either incorrect joins, which lower pairwise precision by merging fragments of different filaments, or refused continuations, which lower pairwise recall by splitting one filament across several instances. Both concentrate at crossings, where endpoint geometry alone can be insufficient to determine ownership. The orientation-based crossing-prevention criterion reduces incorrect joins but cannot eliminate them, since a shallow-angle crossing of two smooth filaments presents locally consistent orientations for the wrong pairing as well as the right one. Bundles that are missing or heavily fragmented in the input mask are a third mode, and no downstream geometry can repair them.

The comparator is one implementation, not the literature. The minimum-turn tracer is a transparent same-mask reference point chosen so that the comparison isolates grouping. It is not a survey of curvilinear tracing methods. The connected-component method serves only as a lower-bound sanity check. These results therefore support a comparison with the specific minimum-turn implementation and settings used here, and the counterexamples in Section 3.1 should be read as evidence that the two approaches have genuinely different failure regimes.

Dependence on input mask quality and scale metadata. FilaSeg cannot recover a filament that is absent from its input, and it may retain or join false axes that the segmenter introduced. Supplying every method the identical mask controls this dependency inside the comparison but does not remove it from practical use. Separately, the parameter rescaling in Section 2.3 requires a physical pixel size. Where acquisition metadata are missing the documented reference scale is used as a fallback, and a wrong scale silently changes the effective meaning of every length-, area- and curvature-valued threshold.

One real development crop, of uncertain independence. All real-data observations derive from a single 512×512 crop from one parent image, annotated once, and that crop informed method development. It is a case study and provides neither a held-out estimate nor a population confidence interval. As noted in Section 2.2.2, the available project records do not establish whether the crop was excluded from the upstream segmenter's training data. Independence between the upstream segmenter and the real evaluation crop is therefore unestablished, and the real result should not be treated as a generalization estimate.

No independent width-accuracy validation. Stage 4 apparent widths are reported as delivered output only. The common-fragment metric is designed for instance identity and is deliberately insensitive to width, so nothing in the primary evaluation constrains width accuracy. A width claim requires matched per-instance reference measurements, which do not exist in this work.

Synthetic-generator assumptions. The generator exposes areal density, centreline-length density, crossings, gaps and intensity variation under exact instance ground truth, which is precisely why it can support a primary endpoint at all. Whether its degradation and appearance distributions resemble those of real upstream segmenter output has not been quantified. The reported confidence intervals characterize repeatability over generator seeds, not transfer to unseen SEM images.

Metric weighting and small development samples. Pairwise fragment statistics weight each instance by roughly the square of the number of fragments it contains, so long filaments split into many fragments influence the score more than short ones. This is defensible for a grouping endpoint but means the metric is not an unweighted per-object average, and it should be read alongside the object-level recovery summary (Section 2.4.2). The sensitivity analyses in Section 3.2 rest on 20 development scenes, with only four scenes at some densities, and the factorial study on four seeds per cell; all are exploratory and cannot support conclusions specific to either density measure.

Missing acquisition and specimen metadata. The SEM instrument and detector, accelerating voltage, working distance, specimen preparation and CNT material identity are not established by the available record, and the relation between the 1536-pixel calibration width and the stored 1534-pixel image width is unconfirmed. These are marked as required actions rather than inferred, and they must be supplied before submission.

Two-dimensional geometry. All inputs are two-dimensional projections. The output is a set of projected centrelines with overlapping instance layers and does not recover three-dimensional over-and-under topology at crossings.

Next validation steps. The results here justify a specific programme rather than a broad claim. In approximate order of value: annotate multiple independent real crops from several parent images, instruments and specimens, by more than one annotator, and seal them before the configuration is fixed, so that inter-annotator agreement bounds the achievable score;

retrain or otherwise establish an upstream segmenter with a documented split that provably excludes those crops, and report end-to-end SEM-to-instance performance with upstream error included; validate Stage 4 apparent widths against matched per-instance reference measurements, independently of the grouping endpoint; quantify how the generator's degradation statistics compare with real segmenter output, and recalibrate the generator where they differ; and extend the comparison to additional published tracing methods on the identical input masks. Until then, an explicit geometric grouping procedure outperforms local minimum-turn continuation on average under the tested synthetic conditions, with the mechanism, failure regimes and representation dependence documented well enough to be checked.

5 Conclusions

FilaSeg reconstructs overlapping filament instances from degraded centreline masks through deterministic Hough vectorization, orientation-aware cleaning and geometric reconnection. On 125 predefined procedural scenes, its Stage-3 centreline output achieved common-fragment F_1 of 0.734 (density-stratified 95% CI [0.722, 0.745]). The paired difference from the fixed minimum-turn comparator was 0.071 (95% CI [0.059, 0.082]).

Which representation is submitted for scoring changes the result without any method output changing (Section 3.1). Equivalent Stage-3 centrelines are the defensible primary grouping representation; Stage-4 centreline changes and width-rendered masks are secondary diagnostics. A secondary comparison on undegraded masks further shows that much of the aggregate advantage over the comparator reflects insensitivity to mask degradation rather than better grouping of intact topology. Automatically selected counterexamples show that the aggregate advantage is not uniform across scenes or densities. The result is therefore evidence for this fixed comparison under the tested generator, not a claim of superiority over filament tracing methods generally.

The single real crop remains a development illustration. Independent real annotations from multiple parent images, complete acquisition and annotation metadata, separate validation of apparent width and other exported measurements, and a citable clean-checkout release are required before broader SEM or quantitative microstructure claims are justified.

CRediT author contribution statement

[To do: Fill in each author's CRediT roles before submission, checked against actual contributions; keep the author names and order as listed above.]

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

The FilaSeg source code, the procedural synthetic benchmark generator, the evaluation subsystem, and the manually annotated real test crop are contained in the accompanying repository. Every numerical result reported here is reproducible from that repository using the commands, environment versions, configuration hashes and seeds recorded in `docs/REPRODUCIBILITY.md`, together with the per-scene evaluation manifest distributed with the manuscript. **[To do: Replace the repository link with a citable archived release (a Zenodo DOI or a**

tagged GitHub release) before submission, and confirm a clean checkout reproduces the reported numbers first – see Section 4.2.]

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